

## ■ Problem 1 (21/06/2021)

Consider a machining cycle to obtain a through hole with a diameter of 40 mm, a depth of 80 mm and a tolerance grade equal to IT7. The machining cycle involves a drilling operation with a twist drill bit with a diameter equal to 37 mm and a reaming operation. The machining operations are carried out on a machine tool with a nominal power equal to 15 kW and an efficiency equal to 90%. Determine:

- the cutting torque for the reaming operation;
- the cutting power for the reaming operation;
- the machining time for the reaming operation;
- Is the reaming operation feasible on the selected machine tool?

### ***Reaming data:***

- Cutting speed:  $v_c = 120$  m/min
- Feed:  $f = 1$  mm/revolution
- Number of cutting edges:  $Z = 8$
- Cutting pressure:  $k_c = 2100$  MPa
- Extra path in and out: 2 mm each



**▪ Solution****a) Cutting torque**

In the reaming operation each cutting edge will remove material with a depth of cut equal to

$$a_p = \frac{D_{final} - D_{drilling}}{2} = \frac{40 - 37}{2} = 1,5 \text{ mm}$$

It follows a chip area before deformation equal to

$$A_D = f_z \cdot a_p = \frac{f}{Z} \cdot a_p = \frac{1}{8} \cdot 1,5 = 0,1875 \text{ mm}^2$$

and a cutting force per each cutting edge equal to

$$F_c = k_c A_D = 2100 \cdot 0,1875 = 393,75 \text{ N}$$

In a reaming operation all the cutting edges are simultaneously working with identical cutting parameters, therefore

$$T_c = Z \cdot F_c \cdot \frac{D_{final} + D_{drilling}}{4} \cdot \frac{1}{1000} = 8 \cdot 393,75 \cdot \frac{40 + 37}{4} \cdot \frac{1}{1000} = 60,64 \text{ Nm}$$



▪ **Solution**

**b) Cutting power**

Therefore, the cutting power is equal to

$$P_c = T_c \omega ; \omega = \frac{2\pi n}{60} = \frac{1000 \cdot v_c}{60 \cdot D/2}$$

$$P_c = T_c \omega$$

$$\omega = \frac{v_c}{D_f/2} \cdot \frac{1000}{60} = \frac{120}{40/2} \cdot \frac{1000}{60} = 100 \frac{\text{rad}}{\text{s}}$$

$$P_c = T_c \omega = 60,64 \cdot 100 = 6.064 \text{ W} = 6,1 \text{ kW}$$

**c) Machining time**

The machining time is equal to

$$t_m = \frac{L_m}{v_f} = \frac{H + e_{in} + e_{out}}{f \cdot n}$$

where

$$n = \frac{1000 \cdot v_c}{\pi D_f} = \frac{1000 \cdot 120}{\pi \cdot 40} = 955 \text{ RPM}$$

Therefore

$$t_m = \frac{80 + 2 + 2}{1 \cdot 955} \cdot 60 = 5,3 \text{ s}$$



- ***Solution***

***d) Is the reaming operation feasible on the selected machine tool?***

The power limit is  $P_{c,limit} = E \cdot P_e = 0,9 \cdot 15 = 13,5$  kW, therefore the reaming operation is feasible.



**■ Problem 2 (15/01/2021)**

Consider a machining cycle to obtain a through hole with a diameter of 30 mm and a depth of 60 mm with a tolerance grade equal to IT8. The machining cycle involves a drilling operation with a twist drill bit with a diameter equal to 28 mm and a finishing operation by means of a single-point cutting tool (boring). The machining operation are carried out on a machine tool with a nominal power equal to 7,5 kW and an efficiency equal to 90%. Determine the cutting force, the cutting power, and the cutting time for both the machining operations. Are the machining operations feasible on the selected machine tool?

**• Drilling data:**

Cutting speed:  $v_c = 290$  m/min

Feed:  $f = 0,10$  mm/revolution

Cutting pressure:  $k_c = 2100$  MPa

Point angle:  $\varepsilon = 160^\circ$

**• Finishing data:**

Cutting speed:  $v_c = 325$  m/min

Feed:  $f = 0,05$  mm/revolution

Single-point cutting tool:  $Z = 1$

Cutting pressure:  $k_c = 2100$  MPa





▪ **Solution**

Twist drilling:

$$A_D = \frac{f \cdot D}{4} = \frac{0,1 \cdot 28}{4} = 0,7 \text{ mm}^2$$

$$F_c = k_c A_D = 2.100 \cdot 0,7 = 1.470 \text{ N}$$

$$T_c = F_c \frac{D/2}{1000} = 1.470 \frac{28/2}{1000} = 20,58 \text{ Nm}$$

$$n = \frac{1000 \cdot v_c}{\pi D} = \frac{1000 \cdot 290}{\pi \cdot 28} = 3.297 \text{ RPM}$$

$$\omega = \frac{2\pi n}{60} = \frac{2\pi \cdot 3.297}{60} = 345,3 \frac{\text{rad}}{\text{s}}$$

$$P_c = T_c \omega = 20,58 \cdot 345,3 = 7.106 \text{ W} = 7,1 \text{ kW}$$

The power limit is  $P_{c,limit} = E \cdot P_e = 0,9 \cdot 7,5 = 6,75 \text{ kW}$ , therefore the drilling operation is not feasible.

In order to determine the cutting time, we need to compute the overall cutting length considering the conical shape of the drill bit end. Therefore:

$$A = \frac{D/2}{\tan \frac{\varepsilon}{2}} = \frac{28/2}{\tan \frac{160^\circ}{2}} = 2,47 \text{ mm}$$

$$L_c = H + A = 60 + 2,47 = 62,47 \text{ mm}$$

$$t_c = \frac{L_c}{v_f} = \frac{L_c}{f \cdot n} = \frac{62,47}{0,1 \cdot 3.297} = 0,1895 \text{ min} = 11,37 \text{ s}$$



**▪ Solution**

*Finishing (boring):*  $Z=1 \rightarrow$  similar to turning operation

This operation is a single-point cutting tool operation (similar to the internal cylindrical turning):

$$a_p = \frac{D_{final} - D_{drilling}}{2} = \frac{30 - 28}{2} = 1 \text{ mm}$$

$$A_D = f \cdot a_p = 0,05 \cdot 1 = 0,05 \text{ mm}^2$$

$$F_c = k_c A_D = 2.100 \cdot 0,05 = 105 \text{ N}$$

$$P_c = F_c v_c = 105 \cdot \frac{325}{60} = 568,75 \text{ W} = 0,57 \text{ kW}$$

The power limit is  $P_{c,limit} = E \cdot P_e = 0,9 \cdot 7,5 = 6,75 \text{ kW}$ , therefore the finishing operation is feasible.

$$n = \frac{1000 \cdot v_c}{\pi D} = \frac{1000 \cdot 325}{\pi \cdot 30} = 3.448 \text{ RPM}$$

$$t_c = \frac{L_c}{v_f} = \frac{H}{f \cdot n} = \frac{60}{0,05 \cdot 3.448} = 0,348 \text{ min} = 20,88 \text{ s}$$

